

Corporate *Power Purchase Agreements*

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Course Director - *Christian E G MSc*



London Petro Academy
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What will this course cover?

The global power markets are rapidly evolving and transitioning towards ever-increasing quantities of clean energy - power purchase agreements allow both IPP power producers and buyers to obtain long-term security to sell and buy renewable energy. For both generators and buyers of renewable energy power purchase agreements need to address their individual circumstances and risks.

This highly specific course aims to introduce the legal, financial and technical structure and hidden pitfalls of PPAs as a means for Sellers and Buyers to minimise their exposure to volatility of price and volume. The structure of physical and financial PPAs will be thoroughly dissected and put into context of the overall risk assessment of the project transaction. We will compare both types of PPAs, their different tariff structure based on either costs or market price and the implication for the seller and buyer.

Delegates will be engaged to discuss recent case studies from around the world, embark on group exercises of risk allocation and will revising real world PPAs clauses in order to dig out the gist of this mission critical project agreement.



Course Objectives

- Leave the course with a comprehensive understanding of the components of PPAs
- Put PPAs in the context of the wider contractual framework
- Learn the best practice when working with PPAs
- A step-by-step guide of the basic types and key features of corporate PPAs
- Analysis case studies and best practice
- Put everything you learn into practice the very next day
- Discover what it takes to make a power purchase agreement “bankable” – i.e. what provisions must a PPA have in order for lenders to provide debt to the project?

Who should attend

- Corporate procurement and Contract managers
- Project developers and Sponsors
- OEM manufacturers, EPC contractors and O&M service providers
- IPPs, Utilities and Grid operators
- Private Equity & Institutional Investors
- Development finance institutions and Commercial lenders
- Legal, Technical and Insurance Advisors
- Government officials and Regulators
- Oil & Gas upstream companies
- Petrochemical and Process industry companies
- Business developers and Sales Managers



Some of our
**Major
Clients**

DAY 1

The corporate PPA in context

- Scale of the corporate PPA market compared to general market size
- Who is buying who is selling what?
- Drivers of corporate PPAs
- Wholesale prices and Interconnection capacity as a mega driver

Basic types and key features of corporate PPAs

- Existing assets versus newly developed assets
- Sleeved and synthetic PPAs
- Contract for difference
- Bundled, certificates only or electricity only PPA
- Behind the meter or 'on-site' PPAs
- Fully contracted asset versus partly merchant asset
- Pooling different load profiles as a buyer group
- Multiple Corporate PPAs and first right of power

Overall structure of a PPA

- Definition of commercial terms
- Risk to bankability
- Obligations and Undertakings of each party
- Conditions Precedents to be on the safe side
- PPA Definitions and Annexes

DAY 2

PPA Timing and Milestones

- From Signing to Expiry Date
- Scheduled Dates, Actual Dates and Events
- Changing Rights and Obligations at each dates
- Release of obligations in certain events
- Pre-COD and Post-COD periods
- Periods to exercise rights, notify and cure
- Termination rights not meeting milestones

Pre-Commercial Operation Date Period

- Order of execution of other project contracts
- Back to back requirements with the EPC contractor
- Development bonds, Performance Bonds and Letter of credits
- Definitions of achieved, declared, contracted and minimum capacities
- Delay and Performance Liquidated Damages – calculation and different structures

Post-Commercial Operation Date Period

- Delivery Point, Metering and Losses – the devil is in the detail
- Invoicing, Billing and Payments procedures or show me the money
- Impact on working capital requirements and bankability
- Availability, Planned Maintenance Outages and requested changes
- Prudent Utility Practice and manufacturer warranties
- Allowed Grid Unavailability Periods, Curtailment and Deemed Energy provisions

DAY3

PPA tariff structures

- Understanding Levelized Cost of Electricity, Cost of Capital and their impact on tariffs
- Tariff Structures for intermittent power generation technologies
- Take-or-Pay versus Take-and-Pay contracts
- The rationale for indexation and escalation of local, foreign inflation and currencies
- Discount to market indexes
- Price collar structures
- Seasonal weighting factors in a PPA
- Locational Marginal Pricing and grid charges

Specific considerations for corporate PPAs

- Forecast wholesale price uncertainty and fixing the strike price
- Counter-party risk and long term creditworthiness of the generator and off-taker
- Power consumption load profile and changes over time
- Additionality and Guarantee of Origin
- Scheduling and balancing risks
- Wholesale power price linked PPA

Events beyond Business-As-Usual

- Dispute Resolution, Cure Periods and Arbitration
- Lenders Step-in Rights
- Political and Natural Force Majeure Events
- Changes of Law and Stabilization clauses
- Consequences of Force Majeure and Changes of Law
- Seller or Buyer Event of Default
- Termination rights of PPA by either side
- Termination payment pricing methodology

This course is available In-House and can be tailored to your specific requirements.

For more information, please contact us on:
+44 (0) 1582 516247

Alternatively, you can email us at
info@londonpetroacademy.co.uk

Due to the global pandemic, all our courses are delivered virtually.

Classroom sessions will resume once the pandemic is over and travel restrictions lifted.

Classroom sessions are organised in different cities around the world.

Question about this course?

Call us on +44 (0) 1582 516247 and one of our representatives will assist you with your enquiry.

3 Easy ways to register

Online: www.londonpetroacademy.co.uk

Email: info@londonpetroacademy.co.uk

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Course Director – Christian E G MSc

Christian is an independent financial advisor with some 18 years' experience in the power & renewable sector.

He started out his career in a leading North European Strategy Consultancy, where he made substantial contributions to build the energy practice within the firm. He developed and implemented innovative distributions concepts for utility clients to face the competitive challenges of the newly liberalized German electricity market.

In 2002, anticipating the tectonic shifts to arrive in the energy sector by renewables, Christian decided to set-up his own advisory boutique to serve exclusively the renewable energy sector. In the following years, Christian, supported world leading Utilities, IPPs and Private Equity Investors, such as Iberdrola, AES, RWE, OMV, Englefield Capital and Hudson Clean Energy Partners to thoroughly assess international market opportunities and successfully assemble project portfolios across mature and emerging European renewable markets. Based on this work his clients were able to realize their investment strategy via the acquisition of local partners, greenfield and operating wind, solar, biomass, geothermal or hydro projects with pipelines of many thousands MW capacity.

Christian became a recommended speaker at leading international energy conferences on topics of market attractiveness and investment drivers for renewables. He formed part of the Working Group 'Financing the Energy Turnaround' in Frankfurt and was one of the first members of

the European Technology Platform for Wind Energy, as well as of EWEA, ACORE, LAWEA, EUBIA, ESHA and EGEC.

Since ten years, Christian, focuses on the particular challenges of renewable energy projects in emerging markets, such as Bulgaria, Turkey, Egypt, Georgia, New Zealand, Taiwan, South Korea, Mexico, South Africa or Pakistan. He flags risks to international investors coming from regulatory environment, grid connection and competitors. Developers are supported to raise equity, mitigates risks via the creation of international project consortia and achieve project finance bankability with domestic and international lenders.

Since five years, Christian has successfully delivered energy finance training to hundreds of executives from more than 80 countries in plus 50 seminars in the UK, Germany, Spain, Italy, Denmark, The Netherlands, UAE, the US, Chile, Mexico, Brazil, South Africa, Georgia, Indonesia and Hong Kong. Additionally he served as a lecturer in the Renewable Energy Finance Studies Program of the Frankfurt School of Finance and Management.

Delegates come from leading financial institutions, industrial corporations and governments such as IFC, FMO, GIZ, IFU, Obviam, KfW, HBOR, European Investment Bank, UNDP, Asian Development Bank, African Development Bank, African Finance Corporation, JBIC, Industrial Development Corporation, Banco Santander, Enel, E.ON, Vattenfall, Endesa, Eskom, Total, Sumitomo, Gamesa, Siemens, Akka Enerji, Akzo Nobel, BP, Shell, Abengoa, Energomontaza, Mitsubishi, SgurrEnergy, PwC, Investec,

RES, The Crown Estate, Coca-Cola, Procter&Gamble, Pacific Gas & Electric, Omrom, OFGEM, Munich Re, AES, Alstom, Schneider Electric, Aveng Group, Air Liquide, NEPRA, Saudi Aramco, King Saud University, Swicorp, Qatar Petroleum, K.A. CARE and many more.

Some of the topics delivered include Renewable Technologies & Economics, Independent Power Projects, Project Finance, Financial Analysis & Structuring, Risk Mitigation, Loan Term Sheets Elements, Power Purchase Agreements, EPC and O&M contracts, Resource Assessments, Government Tenders and Policy Incentives, Electricity Market Structures & Liberalization, Electricity Pricing, Project Transactions & Valuations, Innovative Renewable Business Models.

Due to his versatile experience and personality, Christian has the ability to quickly relate to and leverage different cultural and professional backgrounds of delegates. His objective is not only to train passionately on fascinating energy finance topics, but to deliver a seminar experience, where delegates weave together as a group of persons and exchange not merely business cards, but ideas to spur business beyond.

Christian holds an MSc in Industrial Engineering and Management from the Technical University of Berlin and is a qualified RYA Yachtmaster Ocean Instructor.

He is trilingual in English, Spanish and German and conversational in French.



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Get in touch

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Course Formats

-  Virtual Learning/Live sessions
-  Public/Open/Classroom sessions
-  In-Company/In-House sessions

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